

Sean Kuhlman *Senior Software Engineer*

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SUMMARY

Senior Backend and Platform Engineer with 10+ years of experience building distributed systems, enterprise search platforms, fintech applications, streaming services, and cloud infrastructure. Experienced in Python, Go, TypeScript, Node.js, PostgreSQL, Kafka, AWS, Kubernetes, and Terraform, with recent work applying semantic retrieval and AI-assisted workflows to risk, compliance, and enterprise knowledge systems.

Proven ability to design scalable APIs, modernize cloud platforms, improve production reliability, mentor engineers, and lead technical decisions across product, infrastructure, and security domains.

EXPERIENCE

Senior Software Engineer, Topstep

11/2023 – Present | Lakeville, MN

Fintech platform supporting futures-trader evaluation, funded accounts, account operations, and risk management.

- Design backend capabilities supporting trader-account onboarding, account review, risk evaluation, and operational workflows across Topstep's futures-trading ecosystem.
- Build event-driven services with Kafka and Kubernetes to process trading activity, account-state changes, risk events, and operational alerts across internal systems.
- Develop secure REST APIs that expose account, risk, and trading analytics data to internal operations and compliance applications.
- Created an internal AI-assisted compliance research workflow using FastAPI, LangChain, PostgreSQL, and pgvector to retrieve relevant policies, procedures, and investigation context.
- Designed retrieval pipelines that consolidate compliance documentation and operational knowledge while preserving traceability to source material.
- Strengthen service reliability through Datadog distributed tracing, custom dashboards, structured alerting, and production incident analysis.
- Advance SOC 2 readiness through role-based access controls, audit logging, encryption standards, and cloud access-governance improvements.
- Partner with product managers, compliance specialists, and trader-operations stakeholders to translate complex risk requirements into maintainable platform capabilities.
- Establish code-review, automated-testing, production-readiness, and maintainability standards across backend and supporting application repositories.
- Guide junior engineers through architecture reviews, technical design discussions, debugging, and release-readiness assessments.

Technology: Python, FastAPI, PostgreSQL, pgvector, LangChain, Kafka, AWS, Kubernetes, EKS, DynamoDB, Datadog

Software Engineer, Glean

09/2019 – 08/2023 | Lakeville, MN

Enterprise SaaS platform providing permission-aware workplace search and knowledge discovery.

- Developed Python and Go services for enterprise content ingestion, indexing, search orchestration, metadata processing, and document synchronization.
- Built connector and ingestion workflows that transformed content from enterprise applications into searchable, permission-aware representations.
- Designed APIs supporting search, permissions management, document synchronization, metadata retrieval, and content-access workflows.
- Improved search relevance by refining ranking signals, semantic retrieval strategies, query behavior, and repeatable evaluation processes.
- Contributed to embedding-based retrieval and, during the later portion of the role, LLM-assisted knowledge workflows grounded in enterprise content.
- Modernized ingestion services into containerized cloud-native workloads provisioned through Terraform-managed AWS infrastructure.
- Implemented Datadog and CloudWatch monitoring to improve visibility into ingestion failures, indexing delays, and search-service behavior.
- Partnered with security teams to enforce tenant isolation, source-system permissions, access controls, and auditability across enterprise deployments.
- Introduced RFC-driven technical design practices and mentored four engineers through architecture reviews, debugging, planning, and career development.

Technology: Python, Go, PostgreSQL, Semantic Search, Embeddings, OpenAI APIs, AWS, ECS, Lambda, Terraform, Datadog

Tubi

Software Engineer ||

05/2017 – 08/2019 | Remote

Advertising-supported video-streaming platform supporting content discovery and personalization at scale.

- Enhanced recommendation-metadata pipelines with Node.js, GraphQL, PostgreSQL, and Redis as the content catalog and audience expanded.
- Developed backend services supporting content discovery, personalization, catalog metadata, and downstream distribution workflows.
- Improved API performance through query optimization, caching strategies, and targeted service refactoring.
- Implemented asynchronous content-processing workflows with Amazon SNS and SQS to improve system decoupling and resilience.
- Collaborated with product and analytics teams to deliver improvements to recommendation and content-discovery experiences.
- Contributed to multi-region deployment and operational initiatives that strengthened service availability and release confidence.

Technology: Node.js, GraphQL, Redis, PostgreSQL, AWS SQS, AWS SNS, CloudWatch

Software Engineer I

10/2016 – 04/2017 | Remote

Backend development for customer-facing streaming and account experiences.

- Developed REST APIs supporting authentication, user profiles, saved preferences, and viewing-history functionality.
- Built PostgreSQL queries and backend integrations for customer-facing streaming features.
- Contributed to Docker-based CI/CD automation and repeatable deployment workflows.
- Worked with senior engineers to diagnose production issues and improve application stability and release quality.

Technology: Node.js, PostgreSQL, Docker, REST APIs

Full Stack Developer, Pingdom

12/2015 – 09/2016 | Denver, CO

Website availability and performance-monitoring platform for globally distributed applications.

- Developed React and TypeScript monitoring dashboards for uptime, performance, and infrastructure telemetry visualization.
- Built Django backend services for processing webhook events and monitoring data used by operational reporting workflows.
- Optimized client-side rendering for high-volume time-series datasets and interactive monitoring views.
- Supported alerting, logging, and troubleshooting workflows used by distributed monitoring infrastructure.

Technology: React, TypeScript, Django, PostgreSQL, Webhooks

EDUCATION

Bachelor of Science (BS), Computer Science, *University of Colorado Denver*

2013 – 2016

TOP SKILLS

Backend & APIs

FastAPI, Django, Spring Boot, .NET Core, Node.js, REST APIs, GraphQL APIs, Microservices

Frontend

React, Next.js, TypeScript, JavaScript, GraphQL, HTML5, CSS3

Cloud & DevOps

AWS, Kubernetes, Docker, Terraform, AWS CDK, ECS, EKS, Lambda, CI/CD

Methodologies

Agile, Scrum, RFC Documentation, Architecture Reviews, Technical Mentoring, Code Reviews

Languages

Python, Go, Java, C#, TypeScript, JavaScript, PHP, Ruby, C++

AI & Data Platforms

LangChain, OpenAI APIs, Embeddings, Vector Search, RAG Pipelines, pgvector, Pinecone